

Pre-cooling a superconducting-qubit dilution refrigerator: cascade heat-budget penalty and helium-versus-nitrogen cold-plate coolant selection by CFD

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Abstract

Superconducting quantum processors are refrigerated by dilution refrigerators whose warm pre-cooling stages intercept the radiative and conductive loads arriving from the 300 K environment, at a wall-plug cost that can dominate the energy balance of the computing system. The most consequential warm-end design choice is the first-shield temperature: a helium pulse-tube intercept near 50 K or a liquid-nitrogen intercept at 77 K. This choice is assessed here with a three-dimensional, multi-region conjugate-heat-transfer model of a representative cryostat enclosure—resolving view-factor radiation, the calibrated conduction of a 65-line coaxial harness, and their spatial distribution—coupled to a grid-converged cold-plate heat-exchanger model that ranks the two candidate coolants on a common multi-metric basis. The warmer nitrogen shield imposes no Stage-1 penalty: it intercepts 99.6% of the radiation a 50 K shield intercepts and 6.5% *less* cable conduction. The entire penalty appears one stage down and is conductive: ≈ 92 mW of additional wiring conduction delivered to the 4 K plate (ratio 2.31), worth ≈ 8 W of ideal room-temperature power through the $74\times$ Carnot work multiplier. At the cold plate, nitrogen outperforms helium on every physically decisive metric: $17.9\times$ higher heat-transfer coefficient, 1.36 K wall superheat against 24.4 K, $2.1\times$ lower pumping power, and $40\times$ lower exergy destruction, with a property-based Mouromtseff ratio of 96. For the bare-shield enclosure studied, an ideal-work ledger of both routes favours nitrogen by $\approx 20\%$; real stage

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efficiencies—and multi-layer insulation, which shrinks the Stage-1 base the nitrogen multiplier acts on—reverse this verdict unless the Stage-2 wiring is engineered for low conduction, making the pre-cooling choice inseparable from harness and shield design. The results supply a spatially resolved, benchmark-anchored quantification of the pre-cooling penalty and a transferable, multi-metric framework for cryogenic coolant selection.

Keywords: dilution refrigerator, quantum computing, cryogenics, conjugate heat transfer, pulse-tube refrigerator, liquid nitrogen, exergy, coefficient of performance

1 Nomenclature

2 Roman symbols

A	area (exchange or heated) [m ²]
c_p	specific heat capacity [J kg ⁻¹ K ⁻¹]
D_h	channel hydraulic diameter [m]
f	Darcy friction factor [-]
f_k	area fraction of cable component k [-]
h	convective heat-transfer coefficient [W m ⁻² K ⁻¹]
k	fluid thermal conductivity [W m ⁻¹ K ⁻¹]
$\dot{m}$	coolant mass flow rate [kg s ⁻¹]
Mo	Mouromtseff figure of merit [-]
NTU	number of transfer units [-]
Nu	Nusselt number [-]
p	pressure [Pa]
$\dot{Q}_i$	heat load absorbed at stage i [W]
Re	Reynolds number [-]
$\dot{S}_{\text{gen}}$	entropy generation rate [W K ⁻¹]
T	absolute temperature [K]
T_0	heat-rejection (ambient) temperature [K]
$\dot{W}_i$	wall power charged to stage i [W]
$\dot{W}_p$	pumping power [W]

4 Greek symbols

β	volumetric thermal-expansion coeff. [K ⁻¹]
ε	heat-exchanger effectiveness [-]
ε_i	grey-body surface emissivity [-]
ε_{eff}	effective grey-body emissivity [-]
η_i	stage fraction-of-Carnot efficiency [-]
κ_{eff}	effective harness conductivity [W m ⁻¹ K ⁻¹]
σ	Stefan-Boltzmann constant [W m ⁻² K ⁻⁴]

Subscripts 0 ambient; b bulk; w wall; s shield stage; 4 the 4 K stage.

Acronyms CFD computational fluid dynamics; CHT conjugate heat transfer; COP coefficient of performance; GCI Grid Convergence Index; LN₂ liquid nitrogen; OVC outer vacuum can; PTR pulse-tube refrigerator.

1. Introduction

Superconducting quantum processors operate at 10–20 mK, where thermal noise is attenuated far below the qubit energy gap and coherence can be sustained [1, 2, 3]. Holding this temperature is thermodynamically severe: the Carnot coefficient of performance (COP) at millikelvin temperatures is of order 10^{-5} , real machines achieve a small fraction of it, and a 100-qubit platform dissipating microwatts on chip draws tens of kilowatts at the wall [4, 5, 6]. As processors scale toward the thousand-qubit machines now being engineered [7], the cryogenic plant is increasingly identified as the dominant economic and environmental cost of the technology stack [8, 9, 10].

Most of the heat that this plant must reject never reaches the quantum processor at all: it is intercepted by the warm pre-cooling stages, which stand between the 300 K environment and the 4 K plate of the dilution unit. The principal design choice in this regime is the temperature of the first thermal shield. Modern dry dilution refrigerators use a helium pulse-tube refrigerator (PTR) whose first stage reaches 40–70 K without any liquid-cryogen supply [11]; the historical alternative, a liquid-nitrogen (LN_2) bath at 77.36 K, has long cooled radiation shields in superconducting-magnet cryostats [12, 13, 14] and remains attractive on cost and simplicity grounds [15].

Cryostat design guidance treats the shield choice largely through engineering practice: the design texts synthesise the radiative and conductive budget methods [16, 17], helium-plant studies weigh the refrigeration economics of intercept temperature [15], and millikelvin cryostat reviews document the architectures in service [18, 19]. The thermal consequences of the shield choice for a specific machine have, however, been quantified only by lumped-parameter resistance networks [5, 20, 21]. Such models predict stage temperatures accurately but resolve neither the spatial distribution of the heat flux—where radiation concentrates on the shield, which cable bundles carry the load—nor the active question of how well each candidate fluid actually removes the intercepted heat at the cold plate. Spatially resolved thermal modelling has recently appeared at the component level, notably for high-density signalling harnesses [22], but an enclosure-level conjugate treatment coupling radiation, harness conduction, and shield temperature in a single domain remains absent. Two gaps follow. First, the stage-by-stage passive penalty of raising the first shield from 50 to 77 K has not been quantified by spatially resolved, three-dimensional conjugate heat transfer (CHT). Second, the two candidate fluids have not been ranked as first-stage coolants

on a common, dimensionless, multi-metric basis.

This paper closes both gaps with two coupled computational-fluid-dynamics (CFD) studies. A multi-region CHT model of a representative cryostat enclosure resolves view-factor radiation, calibrated coaxial-harness conduction, and their conjugate coupling on identical geometry for the two shield temperatures, so that the pre-cooling penalty is isolated as the difference between two otherwise-matched solutions. A separate, grid-converged cold-plate heat-exchanger model then compares helium and nitrogen at matched duty through the canonical dimensionless groups, the Mouromtseff figure of merit [23], and entropy-generation analysis [24]. The two studies are joined by a single number—the Stage-1 load the enclosure predicts becomes the duty the cold plate must reject—and their results are expressed in one currency, the ideal room-temperature wall power, through a COP cascade.

2. Models and methods

2.1. Enclosure CHT model

The passive study solves a steady, multi-region conjugate model (`chtMultiRegionSimpleFoam`, OpenFOAM v2412 [25]) of a $0.718 \times 0.718 \times 0.335$ m cryostat section whose scale is consistent with commercial dilution-refrigerator hardware (Fig. 1). Six mesh regions—two near-vacuum gaps and four solid coaxial bundles—are bounded by imposed-temperature surfaces: the warm plate and outer vacuum can (OVC) at the hot-face temperature, the radiation shield and cold plate at the stage temperature. Four cases pair the two shield routes across the two upper spans: Stage 1 ($300 \rightarrow 50$ K He and $300 \rightarrow 77$ K LN₂) and Stage 2 ($50 \rightarrow 4$ and $77 \rightarrow 4$ K).

Each bundle lumps one functional line family of a cabled 65-line control harness—25 drive, 25 flux, 10 readout, and 5 pump lines, following the architecture characterised by Krinner et al. [5]—into a single cylinder with a temperature-dependent effective conductivity

$$\kappa_{\text{eff}} = \frac{1}{T_{\text{hot}} - T_{\text{cold}}} \int_{T_{\text{cold}}}^{T_{\text{hot}}} \left[\sum_k f_k \kappa_k(T) \right] dT, \quad (1)$$

where f_k is the area fraction of cable component k (inner conductor, dielectric, outer shield) and $\kappa_k(T)$ its tabulated cryogenic conductivity [26, 17]. Because cryogenic-alloy conductivity varies by more than an order of magnitude over a stage span, this integral form—implemented as a per-bundle

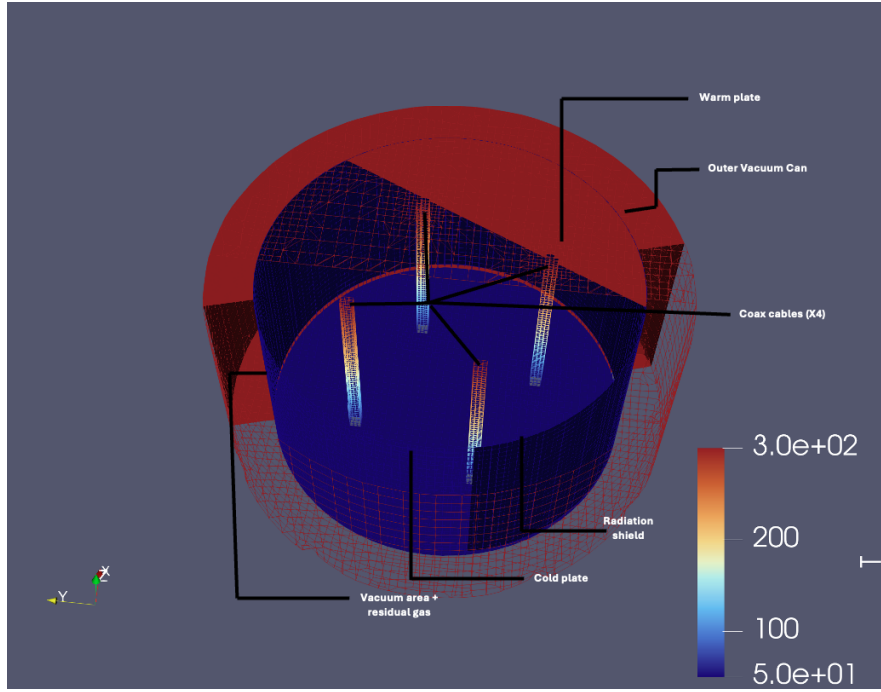


Figure 1: Three-dimensional cutaway of the Stage-1 (50 K) enclosure CHT domain, coloured by temperature. The warm plate (300 K) and outer vacuum can enclose the evacuated gaps and the radiation shield; four lumped coaxial bundles span the 0.335 m height as conduction paths from the warm plate to the cold plate.

fifth-order polynomial with fit error below 0.5%—is what evaluates the conductive load correctly; the calibration reproduces the harness conduction integral independently of the grid. All four families use the stainless-steel reference coaxial construction characterised in Ref. [5]; differentiating the flux family as phosphor bronze is a refinement left for future work, and because both shield routes share the identical harness, the route-to-route ratios on which the conclusions rest are insensitive to this choice.

The four cases share every numerical setting. The segregated SIMPLE algorithm marches the coupled regions to residuals below 10^{-7} ; the enclosing surfaces carry fixed (Dirichlet) temperatures, appropriate for high-conductivity, actively cooled plates; velocity is no-slip with the residual-gas viscosity suppressed, so the gap transport reduces to conduction plus radiation; and the solution is steady-state, the equilibrium heat budget being the quantity of interest. Holding geometry, mesh, and numerics identical between the 50 and 77 K configurations makes the pre-cooling penalty the clean difference between two matched solutions.

Radiative exchange is computed with the `viewFactor` radiosity model. For the near-parallel shield/OVC geometry this is equivalent to the classical grey-body exchange between parallel plates [27], with effective emissivity $\varepsilon_{\text{eff}} = (1/\varepsilon_{\text{OVC}} + 1/\varepsilon_{\text{shield}} - 1)^{-1} \approx 0.0135$, where $\varepsilon_{\text{OVC}} = 0.04$ and $\varepsilon_{\text{shield}} = 0.02$ are the surface emissivities of the polished metals [26, 16]; the closed-form solution provides an exact analytic anchor for validation. The shields are modelled as bare polished surfaces without multi-layer insulation (MLI); the sensitivity of the conclusions to this assumption is assessed in Section 4. The residual gas ($p < 10^{-4}$ mbar) is thermally passive: the mean free path exceeds the gap, bulk convection cannot occur, and the physical free-molecular conduction, which scales linearly with pressure, is bounded below 1% of the stage load by the Kennard relation [28]. The mesh (structured background grid with surface-snapped refinement on the bundles, $0.2\text{--}0.4 \times 10^6$ cells) was run to residuals below 10^{-7} in all regions, and all wall-flux post-processing was audited against closed-form energy balances: conduction against the calibrated $\kappa_{\text{eff}} A/L \Delta T$ integral and radiation against the grey-body anchor. Absolute enclosure loads carry an estimated $\pm 15\%$ uncertainty dominated by emissivity values; mechanism fractions and route-to-route ratios, on which the conclusions rest, are robust to $\pm 3\%$ because both mechanisms scale identically.

A property of the radiative solution deserves emphasis (Fig. 2). The view-factor agglomeration sweep is *non-monotonic*—coarse, production, and

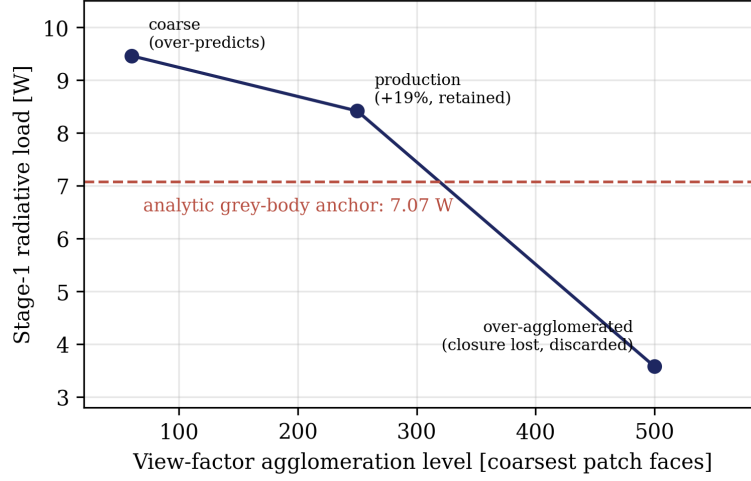


Figure 2: Stage-1 radiative load against view-factor agglomeration level, from the cascade convergence study. The sweep is non-monotonic: the production level sits 19% above the exact grey-body anchor and is retained as a conservative upper bound, while the over-agglomerated level loses radiative closure and is discarded. Richardson extrapolation is therefore inapplicable to the radiative term.

121 over-agglomerated meshes give 9.46, 8.41, and 3.58 W against the 7.07 W ana-
 122 lytic anchor—because heavy face agglomeration degrades view-factor closure
 123 and unphysically suppresses exchange. A standard Richardson extrapolation
 124 is therefore inapplicable to the radiative term, and enclosure radiation is val-
 125 idated against the exact anchor instead, with the production value retained
 126 as a conservative upper bound. Stage 2 corroborates the approach (analytic
 127 3.15 mW versus production 3.44 mW).

128 An external consistency check anchors the model in published hardware
 129 data. Applied to the geometry of the instrumented Bluefors XLD400 char-
 130 acterised by Krinner et al.—aluminium shields of emissivity 0.06 and a first-
 131 shield area of $2\text{--}3\text{ m}^2$ —the same grey-body exchange predicts a first-shield
 132 radiative load of 30–50 W, consistent with the $\approx 50\text{ W}$ those authors report
 133 against a measured first-stage cooling capacity of 30 W at 45 K [5]. The
 134 per-line harness conduction computed here is likewise consistent in magni-
 135 tude with the stainless-steel coaxial estimates of Ref. [5] once the differing
 136 inter-stage spans are accounted for.

2.2. Cold-plate heat-exchanger model

The active study resolves a channelled metal cold plate ($D_h = 10$ mm channels, in the conventional-channel regime of the Kandlikar classification [29], where D_h is the channel hydraulic diameter) rejecting the matched Stage-1 duty of 9.4 W at a Reynolds number $Re = 2300$, the laminar/turbulent edge. The scenario represents a remote-cooling architecture, in which the first-stage duty is removed by a coolant circulated through the shield plate—helium gas from a cryoplant loop or nitrogen from an LN_2 system—rather than by direct conduction to a PTR first stage; distributed circulation loops of this kind are the standard architecture of large-scale cryogenic plants [15] and of the large cryostats now built for quantum platforms [7, 18]. Matching the Reynolds number fixes a common dynamical regime; since any single common basis (equal Re , equal flow rate, equal pumping power) inevitably privileges one fluid, the ranking is cross-checked at matched pumping power and with the geometry- and solver-independent Mouromtseff number in Section 3.2. The two coolants are solved on identical graded meshes with physical models dictated by their density behaviour: LN_2 at 77 K with an incompressible Boussinesq solver (valid, since the product of expansion coefficient and temperature excursion is $\beta\Delta T \approx 0.03$), and helium at 50 K with a compressible, variable-density solver, because its $\beta\Delta T \approx 0.63$ and $\sim 78\%$ density variation place it outside the Boussinesq limit.

Performance is reduced to the canonical groups. The Nusselt number $Nu = hD_h/k$ normalises the heat-transfer coefficient h by the fluid conductivity k ; the Darcy friction factor f and pumping power $\dot{W}_p$ carry the hydraulic cost; the number of transfer units $NTU = hA/(\dot{m}c_p)$ and effectiveness $\varepsilon = 1 - e^{-NTU}$ (with A the heated area, $\dot{m}$ the mass flow rate, and c_p the specific heat) express the exchanger duty in capacity-rate terms [30]; the property-based Mouromtseff figure of merit $Mo = k^{0.6}\rho^{0.8}c_p^{0.4}\mu^{-0.4}$ ranks the coolants independently of geometry and solver [23]; and the entropy-generation rate $\dot{S}_{\text{gen}}$ with its thermal (Bejan) fraction quantifies the second-law cost [24]. The fully developed square-duct values of Shah and London ($Nu_\infty = 3.61$, $f Re = 56.9$) [31] anchor the laminar interpretation.

Four assumptions bound the cold-plate model. The flow is solved laminar at $Re = 2300$, the conservative edge of transition, so the reported heat-transfer coefficients are lower bounds with respect to any turbulence onset; the channel walls are smooth; each coolant’s properties are evaluated by its solver’s thermophysical model rather than held constant; and the solution is steady. The global energy balance of the converged solutions closes to 5.2%

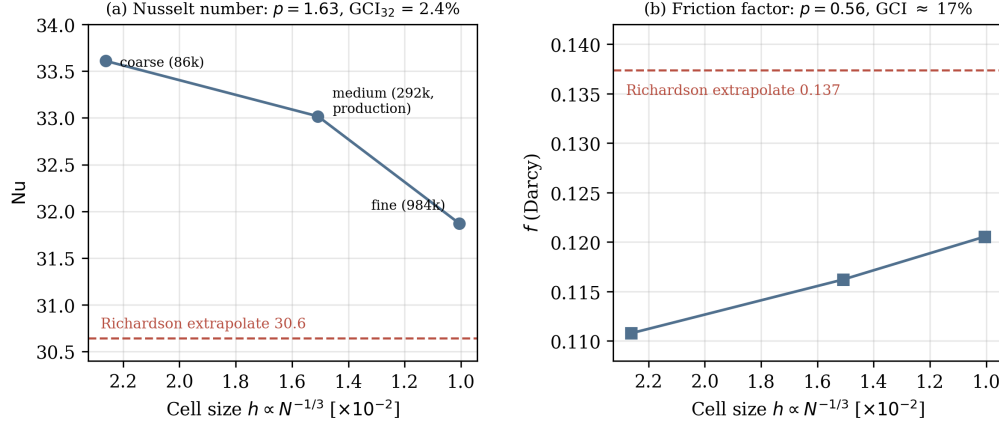


Figure 3: Three-grid convergence of the cold-plate model (LN₂ case, $Re = 2300$), from the grid-independence study. (a) The Nusselt number converges monotonically toward its Richardson extrapolate with a production-mesh GCI of 2.4%. (b) The friction factor converges with low apparent order, as expected of a near-wall quantity; its $\approx 17\%$ GCI is confined to hydraulic quantities that bear on no conclusion.

175 (He) and 2.4% (LN₂), consistent with the discretisation level established
 176 below.

177 Numerical accuracy follows the grid-convergence procedure of Celik et
 178 al. [32], in which the discretisation uncertainty of each quantity is expressed
 179 as a Grid Convergence Index (GCI), on three systematically refined meshes
 180 (86k, 292k, 984k cells; refinement ratio 1.5). Convergence is monotonic for
 181 all reported quantities (Fig. 3): the Nusselt number extrapolates to 30.6 (ap-
 182 parent order 1.63) with a production-mesh GCI of 2.4%, while the friction
 183 factor is grid-sensitive (extrapolated 0.137, $GCI \approx 17\%$), the expected be-
 184 haviour of a near-wall quantity. The friction uncertainty is carried explicitly
 185 into the pressure-drop and pumping-power figures and does not propagate
 186 to any conclusion; the conclusions rest on Nusselt-based and property-based
 187 quantities. Full model dictionaries, assumptions, and the complete metric
 188 set are given in the Supplementary Information.

189 2.3. COP cascade

190 A heat load is meaningful only once referred to the work needed to reject
 191 it. For a stage absorbing $\dot{Q}_i$ at temperature T_i and rejecting to ambient
 192 $T_0 = 300$ K, the ideal wall power is $\dot{W}_i = \dot{Q}_i (T_0 - T_i)/T_i$, with real machines
 193 requiring $1/\eta_i$ times more at stage fraction-of-Carnot efficiency η_i [11, 4].

194 The ideal multiplier $(T_0 - T_i)/T_i$ is 5.0 at 50 K, 2.9 at 77 K, and 74 at 4 K: a
 195 watt displaced from the shield onto the 4 K plate costs an order of magnitude
 196 more at the wall than it saves at the shield. This asymmetry organises all
 197 results below.

198 3. Results

199 3.1. The passive penalty: a Stage-1/Stage-2 mechanism crossover

200 The controlling mechanism changes between the two spans (Fig. 4, Ta-
 201 ble 1). At Stage 1 the 300 K boundary makes radiation dominant, and the
 202 $\sigma(T_{\text{hot}}^4 - T_{\text{cold}}^4)$ flux— σ the Stefan–Boltzmann constant—is set almost en-
 203 tirely by the 300^4 term: the intercepted-radiation ratio between routes is
 204 $(300^4 - 77^4)/(300^4 - 50^4) = 0.996$, a difference of 0.4%. Stage-1 conduction
 205 even mildly *favours* the warmer shield (smaller span): 962 versus 1029 mW,
 206 a ratio of 0.935. The Stage-1 cooling requirement is 8.1 W (analytic-anchored
 207 best estimate: 7.07 W radiation + 1.03 W conduction), rising to 9.4 W on the
 208 conservative production solution. *Stage 1 therefore imposes no meaningful*
 209 *pre-cooling penalty.*

210 At Stage 2 the picture inverts. With the hot face at 50–77 K the radiative
 211 flux has collapsed by two orders of magnitude and conduction governs. The
 212 residual radiation rises steeply between routes, $(77^4 - 4^4)/(50^4 - 4^4) = 5.62$,
 213 but acts on a ~ 3 mW base; the conductive load rises by $2.31\times$ on a 70 mW
 214 base. The Stage-2 budgets are ≈ 74 mW (He route: 70.3 mW conduction
 215 + 3.4 mW radiation) against ≈ 0.18 W (LN₂ route: 162 mW conduction plus
 216 residual radiation), so the warmer shield delivers an extra ≈ 92 mW of *con-*
 217 *duction* to the 4 K plate. All four CFD ratios reproduce their closed-form
 218 benchmarks to better than 0.5%.

219 Spatially, the CHT adds what a lumped network cannot: the view-factor
 220 solution concentrates radiative flux on shield regions with direct sightline
 221 to the OVC, leaving cable penetrations cooler, and the conductive load is
 222 shared unequally among bundles in the ratio of their line counts (25:25:10:
 223 5). In absolute terms, on the 300 $\rightarrow$ 50 K span the drive and flux bundles
 224 carry 396 mW each (38.5% of the harness conduction apiece), the readout
 225 bundle 158 mW, and the pump bundle 79 mW; the same proportions hold on
 226 every span, so on the critical 77 $\rightarrow$ 4 K segment the drive and flux families
 227 deliver 125 of the 162 mW total. This identifies those two bundles as the
 228 dominant conductive paths and therefore the priority targets for intermediate
 229 heat-sinking—a design steer that follows directly from resolving the harness

Table 1: Mechanism decomposition and the 77/50 K first-shield ratios. The analytical column is the closed-form benchmark; CFD and analytic agree to $< 0.5\%$.

Stage	Dominant mechanism	77/50 K (CFD)	Analytical	Interpretation
Stage 1 (radiation)	radiation	0.996	0.996	shield T im- ma- te- rial
Stage 1 (conduction)	conduction	0.935	0.935	warmer shield marginally favourable
Stage 2 (radiation)	small base	5.62	5.62	steep T^4 rise, small ab- so- lute
Stage 2 (conduction)	conduction	2.31	2.31	dominant Stage- 2 penalty

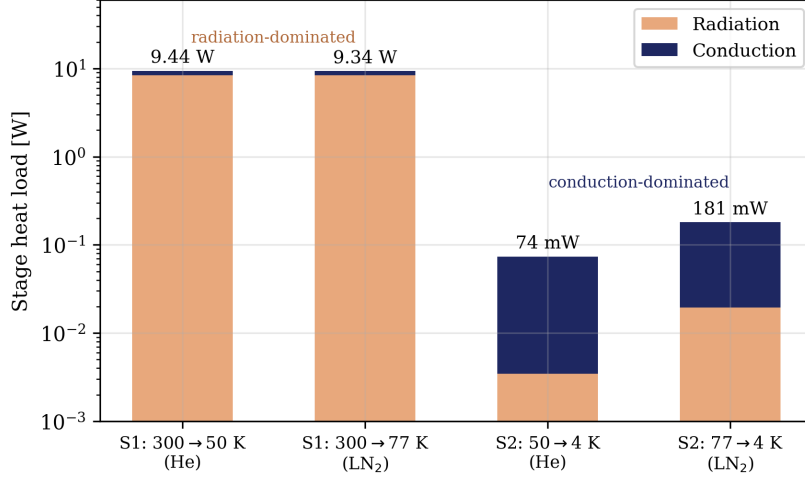


Figure 4: Stage heat loads for the four enclosure cases (production solutions), decomposed by mechanism. Stage 1 is radiation-dominated and nearly identical between routes; Stage 2 is conduction-dominated and is where the two routes diverge—the entire pre-cooling penalty is the extra wiring conduction delivered to the 4 K plate on the 77 K route.

as distinct functional families rather than as a single equivalent conductor.
Figure 5 is reserved for the corresponding field visualisation.

3.2. Active cold-plate performance: nitrogen versus helium

At the matched 9.4 W duty and $Re = 2300$, nitrogen is the superior Stage-1 coolant on every physically decisive axis (Table 2, Fig. 6). It delivers a heat-transfer coefficient $17.9\times$ higher than helium (460.9 versus $25.7 \text{ W m}^{-2}\text{K}^{-1}$) and holds the plate at a wall superheat $T_w - T_b$ (wall minus bulk temperature) of 1.36 K against helium’s 24.4 K—while drawing $2.1\times$ less pumping power. The helium superheat carries an architectural corollary: a plate nominally held at 50 K by forced helium convection floats near 74 K under the full duty—effectively the nitrogen shield temperature—so the nominal 50/77 K distinction survives only where the helium duty is removed conductively by a PTR stage rather than convectively by a circulated gas. Both flows are thermally developing—the Graetz thermal entry length exceeds the channel length by factors of 2.9 (He) and 9.1 (LN₂)—which is why each Nusselt number exceeds the fully developed $Nu_\infty = 3.61$ by a large margin, and why nitrogen, with the longer dimensionless entry region, gains the most from

ParaView cut views to be inserted by the author — (a) mid-plane temperature field of the Stage-1 (50 K) enclosure case, showing the OVC at 300 K, the shield, and the four coax bundles as warm-topped conduction paths; (b) shield-surface radiative heat-flux map (q_r field), showing flux concentration on regions with direct sightline to the OVC and cooler cable penetrations.

Figure 5: Spatial fields of the enclosure solution: (a) mid-plane temperature distribution and (b) shield-surface radiative-flux map, showing the non-uniform loading that a stage-averaged lumped model cannot resolve.

Table 2: Cold-plate heat exchanger at matched 9.4 W duty, $Re = 2300$ (He variable-density, LN_2 Boussinesq). Nu and h carry $GCI \approx 2.4\%$; f and $\dot{W}_p$ carry $GCI \approx 17\%$. The full metric set is given in the Supplementary Information.

Quantity	He (50 K)	LN_2 (77 K)	LN_2/He
Nusselt number Nu ($Nu_\infty = 3.61$)	13.9	33.0	2.4
Heat-transfer coeff. h [$W m^{-2} K^{-1}$]	25.7	460.9	17.9
Wall superheat $T_w - T_b$ [K]	24.4	1.36	0.056
Friction factor f (Darcy)	0.080	0.116	1.46
Pumping power $\dot{W}_p$ [μW]	143	68.5	0.48
NTU / effectiveness ε	0.24/0.21	0.18/0.17	0.77/0.79
Mouromtseff Mo	470	4.5×10^4	96
Exergy destruction [W] ($T_0=300$ K)	15.0	0.37	0.025

247 it. Axial conduction is negligible in both flows (axial-conduction parameter
248 below 6×10^{-4}), justifying its omission from the assessment.

249 Three features of Table 2 warrant care. First, the effectiveness–NTU pair
250 is the one metric that appears to favour helium ($\varepsilon = 0.21$ versus 0.17). This
251 is a capacity-rate artefact, not a contradiction: helium’s very low capacity
252 rate $\dot{m}c_p$ inflates NTU despite its lower h . For a fixed-duty cold plate the
253 figure that matters is how cold the plate stays—the wall superheat—where
254 nitrogen wins by $17.9\times$. Reporting ε alone would invert the engineering
255 conclusion; this is precisely why the assessment is multi-metric. Second, the
256 ranking is robust to coolant basis: at matched pumping power the laminar

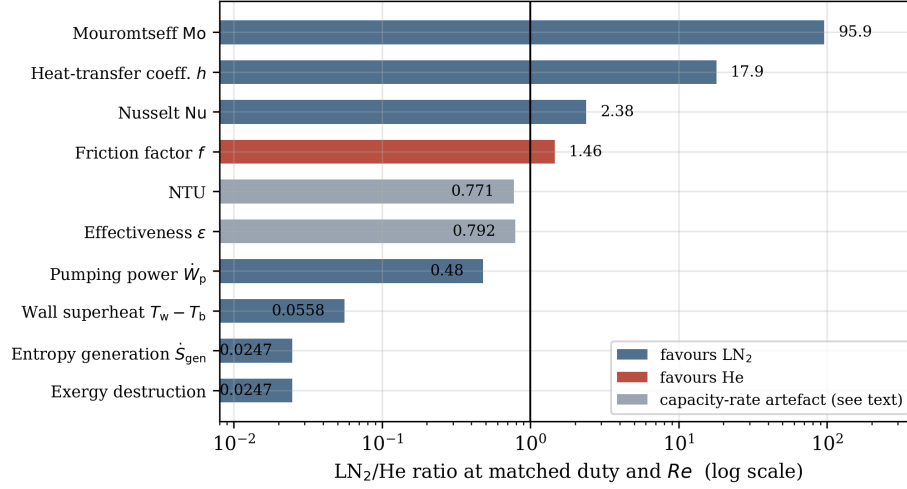


Figure 6: The multi-metric verdict at a glance: LN₂/He ratio for each assessment metric at matched duty and Reynolds number, from the converged CFD solutions. Bars are coloured by which coolant the metric favours; the effectiveness–NTU pair (grey) appears to favour helium but is a capacity-rate artefact, as discussed in the text.

257 scalings leave the advantage set by the thermal-conductivity ratio $k_{\text{LN}_2}/k_{\text{He}} =$
258 7.5, and the property-based Mouromtseff ratio—independent of mesh, solver,
259 and the friction-factor uncertainty entirely—is 96. Third, entropy-generation
260 accounting shows both flows are thermally dominated (thermal share of $\dot{S}_{gen}$
261 above 0.999 in both cases), and nitrogen destroys $40\times$ less exergy (0.37 versus
262 15.0 W) owing to its far lower wall superheat. (The exergy destruction $T_0\dot{S}_{gen}$
263 is referred to the ambient $T_0 = 300\text{ K}$ and can therefore exceed the 9.4 W
264 duty itself: it is the room-temperature work forfeited to the irreversibility of
265 transferring that duty across the wall superheat.) Figure 7 is reserved for
266 the channel-field visualisation.

267 3.3. The combined ledger

268 The two studies close on one statement per route (Fig. 8). On the passive
269 side, nitrogen imposes no Stage-1 penalty and a marginally lower Stage-1
270 load; its entire cost is the extra $\approx 92\text{ mW}$ of Stage-2 conduction, which the
271 $74\times$ multiplier converts to $\approx 6.8\text{ W}$ of ideal wall power ($\approx 8\text{ W}$ including the
272 residual radiative difference). On the active side, nitrogen rejects the Stage-1
273 load far more effectively on every decisive metric. Assembling the ideal-work
274 ledger—Stage-1 loads of 8.1–9.4 W ($\times 5.0$ He, $\times 2.9$ LN₂) plus the Stage-2

ParaView cut views to be inserted by the author — cold-plate channel fields at $Re = 2300$ for the two coolants on the production mesh: (a) LN_2 temperature field on the channel mid-plane, showing the thin thermal boundary layer and 1.36 K wall superheat; (b) He temperature field on the same plane and colour scale, showing the thick boundary layer and 24.4 K superheat; optionally add the velocity magnitude on a transverse cut.

Figure 7: Cold-plate channel temperature fields for LN_2 and He at matched duty and Reynolds number, visualising the $17.9\times$ difference in heat-transfer coefficient as the difference in thermal-boundary-layer thickness and wall superheat.

loads ($\times 74$)—gives totals of 46–53 W for the helium route against 37–41 W for nitrogen: *in ideal-work terms the nitrogen route is the cheaper of the two, by roughly 20%, because intercepting ~ 9 W at a $2.9\times$ multiplier instead of $5.0\times$ saves more than double what the Stage-2 penalty costs.*

4. Discussion

The conventional preference for a colder helium first shield is not an ideal-thermodynamics result. It survives because real machines depart from Carnot unequally across the cascade: writing the stage efficiencies as fractions of Carnot, η_s (shield) and η_4 (4 K), the nitrogen route remains cheaper as long as $\eta_4/\eta_s \gtrsim 0.4$, whereas practical 4 K cryocoolers realise only a few per cent of Carnot while first-stage plants reach an order of magnitude more [11]—so real hardware sits well below the break-even and the helium route prevails in practice. The decisive point is that this break-even is not fixed. Because the Stage-2 penalty is conductive, it is a property of the wiring harness, not of the cryogen: replacing higher-conductivity harness segments with stainless steel or NbTi [26] and heat-sinking the dominant drive and flux bundles identified above compresses the 2.31 conduction ratio toward unity, shrinks the penalty several-fold, and pulls the break-even toward the real-machine ratio. The warm-end conclusion is therefore conditional in a precise and useful way: *a nitrogen-cooled first shield is thermodynamically competitive if, and only if, the Stage-2 wiring is engineered for low conduction.* The choice of cryogen and the design of the harness are not separable decisions—a coupling that

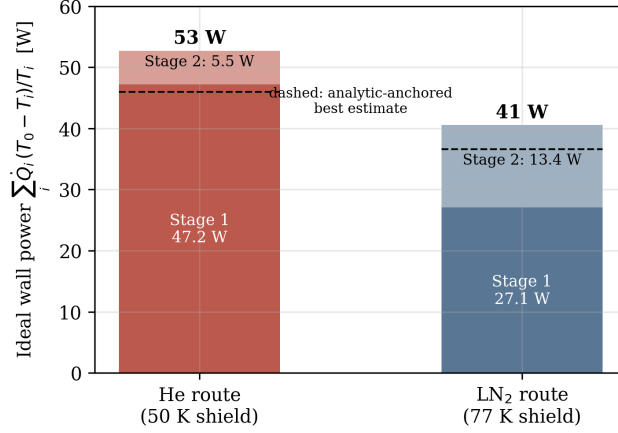


Figure 8: Ideal wall-power ledger of the two pre-cooling routes: stage loads weighted by the Carnot multipliers $(T_0 - T_i)/T_i$ and stacked per route (production solutions; dashed lines mark the analytic-anchored best estimates). The nitrogen route is $\approx 20\%$ cheaper in ideal work despite its Stage-2 penalty.

297 lumped-network treatments, which assign the wiring a fixed resistance, do
 298 not expose.

299 Practical considerations reinforce rather than complicate this framing.
 300 A nitrogen-cooled shield decouples the first-stage duty from the pulse tube:
 301 the PTR can then be sized for the 4 K stage alone, and since mechanical
 302 vibration transmitted from cryocoolers is a recognised concern for precision
 303 cryogenic platforms [21], shifting the largest heat interception to a quiescent
 304 liquid bath is attractive for qubit systems specifically. The price is opera-
 305 tional: an LN₂ inventory to manage, boil-off logistics, and a warmer intercept
 306 for any component anchored to the shield. None of these enters the thermo-
 307 dynamic ledger, but all belong in the machine-level trade study this analysis
 308 is designed to inform, and the helium-plant economics literature provides the
 309 cost framework for it [15].

310 The methodological findings travel beyond this geometry. The demon-
 311 stration that enclosure view-factor radiation does not converge monotonically
 312 under face agglomeration (Fig. 2)—and must be validated against closed-
 313 form grey-body anchors rather than Richardson extrapolation—applies to
 314 radiation-coupled cryogenic CFD generally. The cold-plate comparison il-
 315 lustrates a second transferable lesson: single-metric rankings can invert the
 316 engineering conclusion, as the effectiveness-NTU artefact of Fig. 6 shows,

317 and property-based figures of merit such as the Mouromtseff number pro-
 318 vide a mesh- and solver-independent cross-check on any CFD-based coolant
 319 ranking. The coolant verdict itself is the most transferable quantitative re-
 320 sult, because it rests on fluid properties rather than on the specific channel
 321 geometry.

322 The bare-shield assumption is itself a design variable, and its effect on the
 323 ledger is quantifiable. Wrapping the OVC-facing gap in N layers of MLI sup-
 324 presses the radiative exchange roughly as $1/(N+1)$ [16, 17]: a ten-layer blan-
 325 ket would cut the ≈ 7 W Stage-1 radiation below 1 W, leaving both routes’
 326 Stage-1 budgets conduction-dominated at ≈ 1.6 – 1.7 W and nearly equal. The
 327 nitrogen multiplier advantage (2.9 versus 5.0) then acts on a small base while
 328 the Stage-2 conductive penalty is untouched, and the ideal-work ledger itself
 329 tips to helium (≈ 14 versus ≈ 18 W on the ten-layer estimate). The verdict
 330 of Fig. 8 is therefore specific to lightly shielded enclosures; the transferable
 331 statement is the break-even structure: nitrogen is favoured where Stage-1
 332 loads are large—bare or degraded shields, high optical loading, windowed
 333 cryostats—and helium where MLI has suppressed them, in every case mod-
 334 ulated by the harness-conduction ratio.

335 Three limitations bound the claims. The enclosure’s absolute loads carry
 336 a $\pm 15\%$ band dominated by emissivity uncertainty and are reported as
 337 analytic-to-production ranges; the route-to-route ratios that carry the con-
 338 clusions are robust to $\pm 3\%$. The cold-plate friction factor carries a $\sim 17\%$ GCI
 339 and is confined to quantities that bear on no conclusion. The COP ledger
 340 uses ideal multipliers with stage efficiencies carried parametrically; it ranks
 341 architectures and locates break-evens, but absolute wall powers for a specific
 342 machine require that machine’s measured COP map. Extending the model
 343 below 4 K toward the still and mixing-chamber stages [19, 33, 34], resolving
 344 MLI explicitly rather than through the $1/(N+1)$ scaling, and benchmarking
 345 stage-by-stage against an instrumented refrigerator of the type characterised
 346 in Ref. [5] are the natural next steps.

347 5. Conclusions

348 A three-dimensional conjugate model of a dilution-refrigerator enclosure,
 349 coupled to a grid-converged cold-plate heat-exchanger study, supports the
 350 following conclusions.

- 351 1. The 77 K nitrogen first shield carries no Stage-1 penalty relative to the
 352 50 K helium standard: it intercepts 99.6% of the radiation and 6.5%

- 353 less cable conduction, and the Stage-1 cooling requirement is 8.1–9.4 W
 354 on either route.
- 355 2. The entire pre-cooling penalty appears at Stage 2 and is conductive:
 356 an extra ≈ 92 mW of wiring conduction delivered to the 4 K plate (ra-
 357 tio 2.31), worth ≈ 8 W of ideal wall power through the $74\times$ Carnot
 358 multiplier—concentrated in the drive and flux bundles the spatially
 359 resolved model identifies.
 - 360 3. As a first-stage coolant, nitrogen outperforms helium on every physi-
 361 cally decisive metric: $17.9\times$ higher heat-transfer coefficient, 1.36 versus
 362 24.4 K wall superheat, $2.1\times$ lower pumping power, $40\times$ lower exergy
 363 destruction, and a property-based Mouromtseff ratio of 96; the sole
 364 apparent exception, effectiveness–NTU, is a capacity-rate artefact.
 - 365 4. For the bare-shield enclosure studied, the end-to-end ideal-work ledger
 366 favours nitrogen by $\approx 20\%$ (37–41 versus 46–53 W); real stage efficien-
 367 cies reverse this verdict unless the Stage-2 harness is engineered for
 368 low conduction ($\eta_4/\eta_s \gtrsim 0.4$ break-even), and multi-layer insulation
 369 shifts even the ideal verdict toward helium by shrinking the Stage-1
 370 base. The pre-cooling question is thus reframed: not *which cryogen*,
 371 but *which cryogen given which wiring and which shield*.
 - 372 5. Methodologically, enclosure view-factor radiation does not converge
 373 monotonically under face agglomeration and must be validated against
 374 closed-form grey-body anchors, and single-metric coolant rankings can
 375 invert the engineering conclusion—both lessons transferable to cryo-
 376 genic CFD at large.

377 CRediT authorship contribution statement

378 **Imerson Joao:** Conceptualization, Methodology, Software, Investiga-
 379 tion, Formal analysis, Visualization, Writing – original draft, Writing – re-
 380 view & editing.

381 Declaration of competing interest

382 The author declares that he has no known competing financial interests
 383 or personal relationships that could have appeared to influence the work
 384 reported in this paper.

385 Data availability

386 The OpenFOAM case dictionaries, calibration and gate-check scripts,
387 grid-convergence data, and figure-generation scripts are openly available at
388 <https://github.com/Imerson/Pre-cooling-a-superconducting-qubit-dilution-refrigeration>
389 an archived release will be deposited on Zenodo upon acceptance.

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